

Fereshteh Khodaei

Mechanical Engineer | Dynamics & Control | Robotics | Neural Signal Processing

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SUMMARY

Mechanical Engineer with a PhD in Dynamics, Control, and Vibrations and multidisciplinary research experience in robotics, neural signal processing, computational neuroscience, and mathematical modelling. Experienced in system modelling, control design, neural networks, computational and experimental data analysis, with additional experience in teaching, STEM mentoring, and youth entrepreneurship education. Interested in applying intelligent, adaptive, and data-driven approaches to robotics, autonomous systems, and complex dynamical systems.

RESEARCH INTERESTS

- Dynamic Systems, Control & Optimization
- Robotics, Mechatronics & Flexible Robotic Systems
- System Identification & Mathematical Modeling
- Neural Signal Processing & Computational Neuroscience
- Neural Networks

EDUCATION

K. N. Toosi University of Technology **Tehran, Iran**

PhD in Mechanical Engineering - Applied Design - Dynamics, Control, and Vibrations **Sep 2016 - Sep 2023**

- Graduated with a **GPA of 18.13/20**.
- Thesis Title: "**Dynamic Mapping of Local Field Potential Signals across the Layers of the Primary Visual Cortex in Primate Monkeys**" Supervisors: Prof. S. H. Sadati; Prof. R. Lashgari
- **Notable Courses:** Mathematics (18.5/20) • Neural Networks (20/20) • Advanced Control Systems 1 (18/20) • Advanced Control Systems 2 (17/20) • Nonlinear Control (17.5/20)

Shahrood University of Technology **Shahrood, Iran**

MSc in Applied Mechanical Engineering **Sep 2014 - Sep 2016**

- Graduated with a **GPA of 17.24/20**.
- Thesis Title: "**Modeling and Control of a Robotic Arm Actuated by Twisted String through Discrete Transfer Matrix**" Supervisor: Prof. M. Bamdad
- **Notable Courses:** Continuum Mechanics (20/20) • Kinematics and Dynamics of Robots (17.8/20) • Dynamic Systems (19/20) • Selected Topics in Applied Design - Control with MATLAB/Simulink (19.22/20)

Babol Noshirvani University of Technology **Babol, Iran**

BSc in Solid Mechanical Engineering **Sep 2009 - Sep 2013**

- Graduated with a **GPA of 16.36/20**.
- Thesis Title: "**Modelling and Finite Element Analysis of the 'CHEETA' Flexible Foot**" Supervisor: Prof. M. Dardel
- **Notable Courses:** Physics II (20/20) • Fundamentals of Electrical Engineering I (20/20) • Industrial Drafting (19.75/20) • Numerical Methods (18/20) • Design of Machine and Production Tools (18.1/20) • Mechanism Design (18/20)

National Organization for Developing Exceptional Talents **Gorgan, Iran**

Diploma in Mathematics and Physics Discipline **Sep 2005 - Sep 2009**

- Graduated with a **GPA of 19.47/20**.

RESEARCH EXPERIENCE

K. N. Toosi University of Technology **Tehran, Iran**

PhD Researcher - Neural Signal Processing and Computational Neuroscience **Sep 2016 - Sep 2023**

- Investigated the morphology, polarity, and response dynamics of Local Field Potential (LFP) signals across cortical layers and eccentricities in the primary visual cortex of awake and anesthetized primates.
- Applied time-delayed neural-network and Current Source Density (CSD)-based analyses to study stimulus-dependent information transmission, response timing, amplitude, and layer-specific neural activity.
- Developed interdisciplinary expertise in neural signal processing, computational modelling, experimental neuroscience, neural networks, and dynamical systems, with relevance to brain-machine interfaces and intelligent systems.

IPM Brain Center - Neuroscience and Neuroengineering Center

Tehran, Iran

Graduate Researcher - Neural Signal Processing & Neuroengineering

Sep 2018 - Sep 2021

- Conducted experimental neural data acquisition and analysis, including Local Field Potential (LFP) recordings from the primary visual cortex, focusing on neural signal dynamics and information processing.

Shahrood University of Technology

Shahrood, Iran

Master's Researcher - Modeling and Control of a Robotic Arm Actuated by the Twisted String with Discrete Transfer Matrix

Sep 2014 - Sep 2016

- Developed dynamic models of a twisted-string-actuated robotic arm using the Discrete-Time Transfer Matrix Method (DT-TMM) to characterize system dynamics.
- Designed a PID controller for motion control and analysed the actuator's natural frequencies, mode shapes, and dynamic response.

Babol Noshirvani University of Technology

Babol, Iran

Bachelor's Researcher - Mechanical Analysis of a Cheetah Prosthetic Leg

Sep 2009 - Sep 2013

- Performed finite element analysis of a Cheetah prosthetic leg using ANSYS and SolidWorks to evaluate mechanical stresses, strains, forces, and reaction loads.

PUBLICATIONS

- Khodaei, F., et al., LFP polarity changes across cortical and eccentricity in primary visual cortex. *Frontiers in Neuroscience*, 2023. 17.
- Khodaei, F., et al., Local Field Potential Activity in Response to a Black Square Stimulus in the Primary Visual Cortex. *Advances in Cognitive Science*, 2023.

TEACHING EXPERIENCE

K. N. Toosi University of Technology

Tehran, Iran

Industrial Drawing Instructor 3 Academic Terms Sep 2019 - Jan 2020; Jan 2021 - Jun 2021; Sep 2021 - Jan 2022

- Taught Industrial Drawing to undergraduate students in Mechanical, Electrical, and Materials Engineering, covering technical drawing, engineering graphics, and mechanical design interpretation.

K. N. Toosi University of Technology

Tehran, Iran

Neural Network Teaching Assistant

Jan 2019 - June 2020

- Supported the Neural Networks course by assisting students with concepts, assignments, and MATLAB-based neural-network modelling.

Teen Entrepreneurs Club

Kalaleh, Golestan Province, Iran

Youth Entrepreneurship, Innovation & Creativity Coach

Academic Year 2022-2023

- Delivered an 18-session coaching programme for 27 middle-school girls in an underserved area, promoting creativity, innovation, problem-solving, teamwork, and entrepreneurial thinking through experiential activities; completed Educational Coaching training and received a certificate.

PROFESSIONAL EXPERIENCE

ABCHECK CO (Manufacturer of Agricultural & Industrial Plastic Blown Films)
Systems Analysis & Process Optimization Consultant

Mahmudabad, Iran
June 2025 - Present

- Analyse end-to-end manufacturing processes using process mapping, KPIs, data analysis, and Lean principles to identify bottlenecks, reduce waste, and improve operational efficiency.

SELECTED COURSE PROJECTS

- Chaos control using sliding-mode theory:** Implemented and analysed sliding-mode control for suppressing chaos in nonlinear dynamical systems.
- Static Output-Feedback Control for Vehicle Suspensions: A Single-Step Linear Matrix Inequality Approach:** Implemented LMI-based static output-feedback control for a vehicle suspension system.
- LQG Control Design for Balancing an Inverted Pendulum Mobile Robot:** Applied LQG control to stabilize an inverted-pendulum mobile robot and analysed stability and control performance.
- Active Vehicle Suspensions with Optimal State Feedback Control:** Applied optimal state-feedback control to an active vehicle suspension model, analysing ride comfort and handling performance.
- Artificial Neural Network control of the inverted pendulum:** Implemented and simulated neural-network control for a double-link inverted pendulum.
- Control of the Maya AUV:** Studied dynamic modelling and control of an autonomous underwater vehicle using state-space and classical/modern control approaches.
- Discrete-Time Transfer Matrix Modeling of Flexible Robots under Feedback Control:** Applied DT-TMM to model a flexible robotic arm under feedback control and analysed its dynamic response.

CORE COMPETENCIES

Engineering & Research: Dynamic Systems • Nonlinear & Optimal Control • System Identification • Mathematical Modeling • Robotics & Mechatronics • Flexible Systems • Neural Signal Processing • Neural Networks

Teaching, Mentoring & Leadership: University Teaching • STEM Mentoring • Educational Coaching • Training & Development • Technical Communication • Problem Solving

Software & Tools: MATLAB/Simulink • Python • ANSYS • SolidWorks • CATIA • Arduino • Excel • Power BI • MS Project

CERTIFICATES

Getting Started as a Project Manager: Completion Date: 28 July 2026, Duration: 10 hours 42 minutes

Introduction to Python: Completion Date: 11 November 2025, Duration: 30 hours

Coaching Training Course: Date: 10-13 September 2022, Duration: 30 hours

HSE Workshop: Date: 27-28 February 2019, Duration: 8 hours

Basics of Neuroscience: Date: 26 October-30 November 2017, Duration: 25 hours

Teacher Assistant Workshop: Date: 7 November 2016, Duration: 2 hours

Iso TS 16949:2002 (Certified by Quay School): Date: 15-16 June 2013, Duration: 16 hours

LANGUAGES

English: PTE Academic Overall 60 • Persian: Native • Azeri: Native

REFERENCES

Available upon request